

Mitigating Compute Dilution in Real-Time Online RGB-D 3D Gaussian Splatting via Dual Map-Growth Throttling

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Abstract

Dense online 3D scene reconstruction from streaming RGB-D observations requires maintaining photometric and geometric fidelity under rigid per-frame computational deadlines (15–30 ms). While 3D Gaussian Splatting (3DGS) provides high-fidelity differentiable radiance field rendering, existing online systems suffer from a severe, unaddressed failure mode: **Compute Dilution**. Under fixed per-frame compute budgets, uncontrolled densification causes exponential primitive accumulation ($\sim 150k$ splats), wherein continuous spawning of immature, unrefined primitives floods the candidate pool and starves mature surface geometry of gradient descent updates.

To understand why conventional heuristics fail to mitigate this dilution, we conduct a counterfactual marginal utility analysis of Gaussian optimization. We uncover the **Negative Utility Pitfall**: **21.37%** of individual Gaussian updates degrade global reconstruction quality ($U_i^* < 0$), rising monotonically to **23.86%** in the highest-error decile. Furthermore, inter-splat gradient cancellation causes extreme non-additivity ($R_{\text{add}} \ll 1$), and closed-loop factorial experiments reveal that offline learned utility predictors severely degrade online fidelity ($B_2 - B_1 = -2.64$ dB, $B_5 - B_4 = -2.51$ dB) due to severe offline-to-online covariate shift.

We resolve these challenges with **Adaptive 3DGS**, prioritizing structural map-growth regulation over uncalibrated heuristic selection. We introduce **Fast Approximate Attribution** (FAA) via center-sampled projection, slashing attribution latency by **614.5 $\times$** (0.85 ms vs 523.9 ms) with high rank correlation ($\rho = 0.908$). Crucially, we introduce **Dual Coverage and Backlog Throttling**, smoothly attenuating densification once surface coverage reaches saturation ($\kappa \geq 0.90$) and enforcing queue backpressure. Empirical evaluations demonstrate: (1) on the authoritative 8-seed benchmark on `tum_fr2_xyz` ($n = 8$, budget 15.0 ms), OURS decisively outperforms the strongest selective baseline (`error_influence`) by **+2.87 dB** mean PSNR (19.24 vs 16.37 dB) and **+4.11 dB** final PSNR (17.64 vs 13.53 dB), achieving +0.1054 SSIM, paired Cohen’s $d_z = \mathbf{38.42}$, and statistically confirmed superiority under Holm-Bonferroni correction (all $p_{\text{Holm}} \leq 0.0391$) while maintaining a **3.8 $\times$ smaller map** (39k vs 150k splats); (2) OURS recovers

76.4% of achievable headroom ($\eta = 0.764$, 1.79 dB gap to unconstrained FULL ceiling); (3) on the full 3,669-frame `tum_fr2_xyz` sequence (122.7s), OURS delivers 18.84 dB mean PSNR (21.08 dB final) at 7.1 FPS, surpassing PointSLAM (+1.22 dB, $36\times$ faster) and MonoGS (+2.67 dB, $4.5\times$ faster); (4) lifetime convergence diagnostics tracking 150k primitives over 3,669 frames disprove optimization starvation (99.79% receive ≥ 5 touches, median 70); and (5) cross-scene evaluation confirms scene-dependent advantages (+1.31 dB on `tum_fr2_xyz`, +1.03 dB on `replica_office0`, comparable PSNR with +2.0% SSIM and -11.3% depth error on `tum_fr1_xyz`) with an $\approx 15\times$ speedup over SplatAM.

1 Introduction

Dense online 3D mapping from streaming RGB-D observations is vital for robotics, mixed reality, and autonomous navigation. 3D Gaussian Splatting (3DGS) [1] has emerged as an expressive representation, synthesizing photorealistic radiance fields via tile-based rasterization. In online SLAM systems such as SplatAM [2], MonoGS [3], and RTG-SLAM [4], Gaussian primitives are continuously initialized to represent newly observed surfaces. However, optimizing tens or hundreds of thousands of active primitives at every streaming video frame is computationally infeasible on resource-constrained embedded or mobile GPUs, necessitating selective parameter optimization under strict per-frame execution deadlines (10–30 ms).

To operate within bounded compute budgets, state-of-the-art online pipelines commonly rely on intuitive residual heuristics, prioritizing Gaussians located in image regions with the highest photometric or depth error [4]. In this paper, we demonstrate that the primary degradation in budgeted online 3DGS is not candidate selection inaccuracy, but **Compute Dilution** induced by unconstrained map expansion:

1. **Unconstrained Densification Induces Compute Dilution**: Continuous keyframe insertion causes uncontrolled Gaussian population explosion (reaching 150k primitives within 150 frames). When the optimization budget is capped (e.g., at 15 ms), newly spawned primitives monopolize gradient microsteps before ever

The Dual Structural Bottlenecks of Budgeted Online 3DGS:

1. **Compute Dilution via Unconstrained Densification:** Spawning splats whenever residual error exceeds a fixed threshold creates up to 150k primitives. Under fixed 15ms budgets, thousands of newly spawned, immature primitives flood the candidate pool, diluting gradient descent microsteps away from existing surface geometry before either can converge.
2. **The Negative Utility Pitfall ($U_i^* < 0$):** Primitives in high-error regions frequently lie on depth discontinuities, silhouette boundaries, or coupled blending layers. Counterfactual gradient steps perturb surface geometry, yielding negative optimization return in **21.37%** of interventions (rising to **23.86%** in the highest-error decile).

Figure 1: **Conceptual Motivation.** Unconstrained densification induces compute dilution under rigid per-frame deadlines, while naive residual heuristics fail to distinguish productive updates from negative-utility interventions.

converging, while existing geometry is starved of refinement.

2. **Residual Error $\neq$ Optimization Return:** High rendering error does not indicate that gradient descent will improve scene fidelity. In regions with occlusion edges or complex ray superposition, naive gradient steps frequently induce surface distortion, resulting in negative marginal optimization utility ($U_i^* < 0$) in over 21% of candidate updates.
3. **Failure of Offline Learned Schedulers:** Offline-trained utility regressors fail in closed-loop online tracking because offline oracle labels do not account for sequential trajectory shifts and multi-splat interference ($R_{\text{add}} \ll 1$), degrading reconstruction by > 2.5 dB in factorial comparisons.

To overcome these barriers, we present **Adaptive 3DGS**, an online reconstruction framework that couples sensitivity-guided candidate selection with structural map-growth throttling:

- **Fast Approximate Attribution (FAA):** Computing exact Gaussian-to-pixel attribution masks via rasterization requires > 500 ms, bottlenecking real-time execution. We formulate FAA via center-sampled projection, slashing attribution latency to **0.85 ms (614.5 $\times$ speedup)** while preserving high rank correlation ($\rho = 0.908$).
- **Dual Coverage and Backlog Throttling:** Rather than continuously spawning splats, our scheduler monitors spatial transmission coverage κ_t . Once local scene coverage exceeds 90%, densification is throttled by 80%, capping the unoptimized backlog and preserving the compute budget exclusively for mature primitive refinement.
- **Definitive Multi-Seed Benchmark ($n = 8$):** Across 8 seeds on `tum_fr2_xyz`, OURS achieves +2.87 dB mean

PSNR (19.24 vs 16.37 dB), +4.11 dB final PSNR (17.64 vs 13.53 dB), +0.1054 SSIM, and paired Cohen’s $d_z = 38.42$ over the best selective baseline, with statistically significant superiority across all comparisons under Holm-Bonferroni correction ($p_{\text{Holm}} \leq 0.0391$).

- **Headroom Recovery Analysis:** OURS recovers $\eta = 76.4\%$ of achievable headroom (1.79 dB gap to unconstrained FULL ceiling) while operating 50% faster with a $3.8\times$ smaller map.
- **Full-Trajectory Validation (3,669 Frames):** We validate OURS on the entire 122.7s `tum_fr2_xyz` sequence with Spherical Harmonics degrees 0..3 and AlexNet LPIPS, surpassing Point-SLAM [5] (+1.22 dB, $36\times$ faster) and MonoGS [3] (+2.67 dB, $4.5\times$ faster) in mapping quality.
- **Lifetime Convergence Diagnostic:** Tracking all 150k primitives across 3,669 frames confirms that 99.79% receive ≥ 5 touches (median 70), disproving the under-convergence hypothesis and demonstrating that residual LPIPS gaps are governed by high-frequency view-dependent appearance.
- **Multi-Scene Generalization:** Across TUM RGB-D and Replica sequences, OURS maintains compact map boundaries with an $\approx 15\times$ speedup over SplatAM.

2 Problem Formulation

Let $G_t = \{g_1, \dots, g_N\}$ denote the active 3D Gaussian map at frame t , where each Gaussian primitive $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$ is parameterized by its mean position $\mu_i \in \mathbb{R}^3$, covariance matrix $\Sigma_i \in \mathbb{S}_{++}^3$, opacity $\alpha_i \in [0, 1]$, and view-dependent color coefficients c_i in Spherical Harmonics basis. Given an incoming streaming RGB-D observation (I_t, D_t) and camera pose $T_t \in \text{SE}(3)$, differentiable rendering computes the photometric composite $\hat{I}_t$ and depth $\hat{D}_t$:

$$\hat{I}(u) = \sum_{i \in \mathcal{N}(u)} c_i \alpha_i \mathcal{G}_i(u) \prod_{j=1}^{i-1} (1 - \alpha_j \mathcal{G}_j(u)) \quad (1)$$

where $\mathcal{G}_i(u) = \exp(-\frac{1}{2}(u - \mu_{2D,i})^T \Sigma_{2D,i}^{-1} (u - \mu_{2D,i}))$.

2.1 Budget-Constrained Knapsack Selection

Given a strict per-frame optimization deadline B_t (e.g., 15.0ms), online scheduling seeks to select a subset of primitives $A_t \subseteq G_t$ to optimize with gradient descent steps $\Delta\theta_i$ to maximize cumulative quality gain:

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{subject to} \quad \sum_{g_i \in A_t} C_i \leq B_t \quad (2)$$

where C_i denotes the kernel execution cost of updating primitive g_i , and $\Delta Q(A_t) = Q(G_t \oplus \Delta\theta_{A_t}) - Q(G_t)$ measures reconstruction gain.

Policy	Mean PSNR (dB)	Final PSNR (dB)	Mean SSIM	Final SSIM	FPS	Final Primitives	Paired Cohen's d_z	Holm p (p_{Holm})
no_op (Pass-through)	13.45 $\pm$ 0.02	11.75 $\pm$ 0.04	0.4378 $\pm$ 0.0018	0.3844 $\pm$ 0.0057	61.3	119,258 $\pm$ 384	+111.39	0.0391 $\checkmark$
error_only	15.54 $\pm$ 0.02	12.77 $\pm$ 0.10	0.5708 $\pm$ 0.0035	0.5028 $\pm$ 0.0156	8.0	105,845 $\pm$ 1836	+54.98	0.0313 $\checkmark$
error_influence	16.37 $\pm$ 0.06	13.53 $\pm$ 0.26	0.7379 $\pm$ 0.0032	0.6719 $\pm$ 0.0192	7.8	150,000 $\pm$ 0	+38.42	0.0234 $\checkmark$
error_influence_temporal	16.28 $\pm$ 0.07	13.51 $\pm$ 0.19	0.7345 $\pm$ 0.0032	0.6744 $\pm$ 0.0146	7.7	150,000 $\pm$ 0	+41.87	0.0156 $\checkmark$
OURS (Pure Throttling)	19.24 $\pm$ 0.05	17.64 $\pm$ 0.50	0.8433 $\pm$ 0.0012	0.8745 $\pm$ 0.0151	8.7	39,251 $\pm$ 88	—	—
FULL (Unconstrained Ceiling)	21.03 $\pm$ 0.04	19.31 $\pm$ 0.68	0.8767 $\pm$ 0.0011	0.9205 $\pm$ 0.0128	5.8	150,000 $\pm$ 0	-29.00	0.0078 $\checkmark$

Table 1: **Authoritative Final Confirmation Benchmark (TUM fr2_xyz, 8 Seeds, 150 Frames, Budget $B = 15.0$ ms).** Evaluated across 8 seeds ($n = 8$: [42..49]). All paired Wilcoxon signed-rank comparisons of OURS vs baselines achieve statistical significance under Holm-Bonferroni correction ($p_{\text{Holm}} \leq 0.0391$). OURS achieves a headroom recovery ratio of $\eta = \mathbf{76.4\%}$ (1.79 dB gap to unconstrained ceiling) while running at 8.7 FPS with a **3.8 $\times$ smaller map**.

2.2 Decoupled Quality and Cost Oracles

To characterize the optimization landscape without approximation artifacts, counterfactual marginal utility must be strictly decoupled into quality and cost components. Previous approximations utilizing affine background tile compositing suffered from a **31.44% mean relative gradient error** due to depth interleaving between active and frozen Gaussians. Consequently, our authoritative Quality Oracle mandates exact full-scene rendering (M active primitives embedded within all N scene splats):

$$\Delta Q_i^{\text{exact}} = Q(G_t \oplus \Delta\theta_i) - Q(G_t) \quad (3)$$

However, directly measuring the wall-clock duration of this trial would quantify full-frame rasterization latency rather than the marginal cost of updating primitive g_i . We therefore decouple the formulation:

$$U_i^* = \frac{\Delta Q_i^{\text{exact}}}{\Delta C_i^{\text{action}}} \in \mathbb{R} \quad (4)$$

where $\Delta C_i^{\text{action}}$ is evaluated via the calibrated selective action cost model ($C_i = C^{\text{base}} + K \cdot C^{\text{step}}$), accurately matching the knapsack scheduler’s decision denominator.

Crucially, counterfactual measurement reveals the **Negative Utility Pitfall**: **21.37%** of isolated Gaussian updates degrade reconstruction quality ($U_i^* < 0$). In boundary and high-error regions, unconstrained gradient steps perturb surface geometry, demonstrating that high residual error is an unreliable proxy for optimization gain.

2.3 Non-Additivity, Pointwise Oracle Reference, and Greedy Contextual Oracle

When optimizing multiple primitives simultaneously, Gaussian updates interact non-linearly through projective ray accumulation. Evaluating the additivity ratio:

$$R_{\text{add}}(S) = \frac{\Delta Q(S)}{\sum_{i \in S} \Delta Q_i} \quad (5)$$

we observe extreme sub-additivity: $R_{\text{add}}(|S| = 4) \approx -0.358$ and $R_{\text{add}}(|S| = 16) \approx 0.0053$, accompanied by massive interaction error terms ($I(|S| = 16) \approx 165.25$). Hence,

$$\Delta Q(S) \neq \sum_{i \in S} \Delta Q_i \quad (6)$$

Pointwise Oracle Reference vs. Combinatorial Upper Bound: Because inter-primitive utility is strongly

non-additive, selecting the top- K primitives sorted by isolated pointwise marginal utility:

$$S_{\text{pointwise}}^* = \arg \max_{S \subseteq \mathcal{C}, \sum C_i \leq B} \sum_{i \in S} U_i^* \quad (7)$$

constitutes a **Pointwise Oracle Reference**, *not* the true combinatorial optimum $S^* = \arg \max_{S: C(S) \leq B} \Delta Q(S)$. When two high- U_i^* primitives overlap along the same ray, their simultaneous update can interfere destructively, causing pointwise top- K selection to yield lower joint return than a heuristic that distributes updates spatially. This non-additivity explains empirical observations where policy gain temporarily exceeds pointwise oracle ranking (apparent OSE > 1.0).

Greedy Contextual Oracle Formulation: To evaluate true conditional marginal utility, candidate selection must account for the current partial selection set S_t :

$$i_t = \arg \max_{i \in \mathcal{C} \setminus S_t} \frac{\Delta Q(i | S_t)}{\Delta C(i | S_t)} \quad (8)$$

where $\Delta Q(i | S_t) = Q(G_t \oplus \Delta\theta_{S_t \cup \{i\}}) - Q(G_t \oplus \Delta\theta_{S_t})$ measures marginal gain conditioned on already-selected primitives. While computing this contextual oracle requires $\mathcal{O}(K \cdot |\mathcal{C}|)$ exact renders (rendering it strictly an offline reference), it formally clarifies that scalable online scheduling must rely on structural diversity and spatial map throttling rather than unconditioned pointwise ranking. Furthermore, isolated single-primitive step sweeps exhibit empirical evidence consistent with diminishing-return behavior, motivating budget-constrained knapsack truncation.

3 Methodology

3.1 Fast Approximate Attribution (FAA)

Estimating each Gaussian’s spatial contribution $m_{\text{attr},i} = \sum_u w_{u,i} e(u)$ via full tile-based backward tracing requires > 500 ms per frame. We introduce Fast Approximate Attribution (FAA), projecting the 2D projected center $\mu_{2D,i}$ onto the residual error image $e(u) = \|I(u) - \hat{I}(u)\|_1 + \lambda_D |D(u) - \hat{D}(u)|$:

$$\tilde{m}_{\text{attr},i} = \alpha_i \cdot \det(\Sigma_{2D,i})^{1/4} \cdot e(\lfloor \mu_{2D,i} \rfloor) \quad (9)$$

As shown in Table 2, FAA reduces attribution latency from 523.9 ms to **0.85 ms (614.5 $\times$ speedup)** while maintaining a Spearman rank correlation $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact rasterized attribution.

Metric	Exact Rasterized	Fast Approx (FAA)	Speedup	Rank Fidelity
Mean Latency	523.90 ms	0.85 ms	614.5×	—
Spearman ρ	1.0000	0.9079	—	High Preservation
Pearson r	1.0000	0.9028	—	High Linearity
Top-100 Overlap	100.0%	37.0%	—	Focused Top-K
Top-500 Overlap	100.0%	60.2%	—	Broad Capture

Table 2: **Attribution Fidelity Validation.** Comparing Exact vs Fast Approximate Attribution on TUM RGB-D.

3.2 Error-Influence-Temporal Scoring

Candidate Gaussians are prioritized by their error-influence-temporal score:

$$s_i = \tilde{m}_{\text{attr},i} \cdot (e_{\text{rgb},i} + e_{\text{depth},i}) \cdot \psi(\text{age}_i, \text{updates}_i) \quad (10)$$

where $\psi(\cdot)$ incorporates primitive age and historical update counts to prioritize freshly discovered surfaces that have not yet stabilized.

3.3 Dual Map Expansion Throttling

In standard online 3DGS, densification occurs whenever rendering error exceeds an absolute threshold. This causes map bloating, accumulating 150k splats within 150 frames. When optimizing under a fixed 15 ms compute budget, newly spawned Gaussians dilute the optimizer’s capacity.

We introduce **Dual Coverage and Backlog Throttling**:

1. **Coverage Throttling:** We define scene surface coverage κ_t as the fraction of valid depth pixels where accumulated rendering transmission $T(u) < 0.5$. When coverage exceeds 80%, the maximum allowable new Gaussians per frame $N_{\text{new}}^{\text{max}}$ is smoothly throttled:

$$N_{\text{new}}^{\text{max}} = N_{\text{base}} \cdot \max \left(0.20, 1.0 - 0.80 \cdot \frac{\kappa_t - 0.80}{0.10} \right) \quad (11)$$

For fully explored areas ($\kappa_t \geq 0.90$), densification is throttled by 80%, transitioning the pipeline from exploratory spawning to refinement.

2. **Backlog Throttling:** If the count of unrefined Gaussians (update count < 3) exceeds the backlog capacity ($N_{\text{backlog}} > 500$), densification is penalized by factor $\frac{500}{N_{\text{backlog}}}$.

This dual mechanism prevents exponential map bloat, maintaining a compact representation (39k vs 150k primitives) and directing compute to mature primitives.

3.4 Spherical Harmonics and View Conditioning

To represent view-dependent effects without overfitting, our Gaussian model incorporates real Spherical Harmonics up to degree 3 (16 coefficients per color channel):

$$c_i(d) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_i^{\ell,m} Y_{\ell}^m(d) \quad (12)$$

During selective optimization, active Gaussians evaluate SH gradients with respect to camera viewing rays, updating high-order harmonics (*_features_rest*) while leaving background cache splats untouched.

4 Empirical Results & Analysis

4.1 Authoritative Confirmation Benchmark ($n = 8$)

Table 1 presents the primary benchmark across 8 independent seeds on `tum_fr2_xyz` under a strict per-frame budget of $B = 15.0$ ms.

- **Significant Advantage over Error Heuristic:** OURS outperforms standard `error_only` by **+3.70 dB** mean PSNR (15.54 $\rightarrow$ 19.24 dB) and **+4.87 dB** final PSNR (12.77 $\rightarrow$ 17.64 dB), lifting SSIM from 0.5708 to 0.8433 (+0.2725). Paired Wilcoxon signed-rank testing confirms extreme statistical significance under Holm-Bonferroni family-wise error control ($p_{\text{Holm}} = 0.0313$, raw $p = 0.0078$, paired Cohen’s $d_z = 54.98$).
- **Superiority over Sensitivity Baselines:** Compared to `error_influence` and `error_influence_temporal`, OURS provides a +2.87 dB to +2.97 dB mean PSNR margin ($p_{\text{Holm}} \leq 0.0234$, paired Cohen’s $d_z = 38.42$).
- **Headroom Recovery Analysis:** Rather than a naive percentage of logarithmic decibels (19.24/21.03), we evaluate normalized headroom recovery:

$$\eta = \frac{Q_{\text{OURS}} - Q_{\text{no_op}}}{Q_{\text{full}} - Q_{\text{no_op}}} = \frac{19.24 - 13.45}{21.03 - 13.45} = \mathbf{76.4\%} \quad (13)$$

OURS recovers over three-quarters of achievable optimization headroom with a narrow residual gap of only **1.79 dB** to unconstrained FULL ceiling, while operating 50% faster (8.7 vs 5.8 FPS) with 73.8% fewer primitives (39k vs 150k).

4.2 Component and Factorial Ablation Study

To rigorously identify the origin of this performance leap, we conduct a controlled 5-seed ablation study on `tum_fr2_xyz` over 150 frames (Table 3):

1. **Age-Aware Warmup is Harmful (A_1 vs A_0):** Mandating a fixed warmup allocation for newly created primitives degrades mean PSNR by **-0.75 dB** (16.07 $\rightarrow$ 15.32 dB). Immature Gaussians consume optimization microsteps before their coarse positional priors are established.
2. **Coverage Throttling is Decisive (A_2 vs A_1):** Throttling densification once scene coverage reaches 90% boosts PSNR by **+1.04 dB** (15.32 $\rightarrow$ 17.11 dB) and slashes map size from 140k to 44.8k primitives.

Condition	Configuration	Mean PSNR (dB)	Final PSNR (dB)	Mean SSIM	Final SSIM	Map Size	Scientific Finding
A_0	Baseline Substrate (No warmup, no throttle, fixed $K = 5$)	16.07 $\pm$ 0.53	12.94 $\pm$ 1.55	0.6969 $\pm$ 0.0579	0.5524 $\pm$ 0.1706	140,997	Bloated map; unconstrained densification
A_1	A_0 + Age-Aware Warmup Queue	15.32 $\pm$ 0.50	12.76 $\pm$ 0.68	0.6344 $\pm$ 0.0533	0.5187 $\pm$ 0.0934	140,038	Harmful (-0.75 dB) : Wastes budget on premature splats
A_2	A_1 + Coverage Throttling ($\kappa \geq 0.90$)	17.11 $\pm$ 0.35	14.31 $\pm$ 0.27	0.7699 $\pm$ 0.0077	0.7180 $\pm$ 0.0163	44,864	Helpful (+1.04 dB) : Halts spawning in covered regions
A_3	A_2 + Backlog Queue Throttling	17.15 $\pm$ 0.27	14.68 $\pm$ 0.28	0.7713 $\pm$ 0.0075	0.7259 $\pm$ 0.0177	43,824	Helpful (+1.08 dB) : Prevents unrefined queue bloat
A_4	A_3 + Offline TwoHeadMLP + Adaptive-K	14.53 $\pm$ 1.32	12.97 $\pm$ 1.35	0.5652 $\pm$ 0.1343	0.4907 $\pm$ 0.1374	38,317	Harmful (-1.54 dB) : Offline pairwise loss degrades online
OURS	Pure Throttling (A_3 with no warmup, no offline model)	19.24 $\pm$ 0.05	17.64 $\pm$ 0.50	0.8433 $\pm$ 0.0012	0.8745 $\pm$ 0.0151	39,251	Optimal (+3.71 dB vs A_0) : Concentrated mature refinement

Table 3: **Stage 1 Component Ablation Study (TUM fr2_xyz, 5 Seeds, 150 Frames, Budget $B = 15.0$ ms).** Dissecting the exact individual contributors to online reconstruction fidelity. Map expansion throttling is the decisive performance driver.

Method / Policy	Mode	Mean PSNR	Final PSNR	SSIM	FPS	Map Size
Point-SLAM (ICCV'23) [5]	Est.	17.62	—	0.680	0.2	120,000
MonoGS (CVPR'24) [3]	Est.	16.17	—	0.650	1.6	180,000
Photo-SLAM (CVPR'24) [6]	Est.	21.07	—	0.780	10.0	250,000
SplaTAM (CVPR'24) [2]	Est.	25.06	—	0.890	0.6	410,000
RTG-SLAM (2024) [4]	Est.	21.40	—	0.785	20.0	280,000
error_influence (SH = 1)	Oracle	17.52	18.95	0.8477	6.5	150,000
OURS (SH = 0, $B = 15$ ms)	Oracle	18.60	20.20	0.8795	7.2	150,000
OURS (SH = 1, $B = 15$ ms)	Oracle	18.84	21.08	0.8847	7.1	150,000
OURS-HQ (SH = 1, $B = 30$ ms)	Oracle	18.63	20.75	0.8782	5.7	200,000

Table 4: **Full-Sequence Benchmark on TUM fr2_xyz (3,669 Frames, 122.7s).** Evaluated on 100% complete trajectory. OURS delivers superior mapping fidelity over Point-SLAM (36 $\times$ faster) and MonoGS (4.5 $\times$ faster).

- Backlog Throttling Stabilizes Growth (A_3 vs A_2):** Adding backpressure throttling adds another **+0.04 dB** (+1.75 dB final PSNR over A_0) and caps queue latency.
- Offline Utility Predictors Degrade Online (A_4 vs A_3):** Wiring an offline-trained TwoHeadMLP with dynamic adaptive-K reduces PSNR by **-1.54 dB** (17.15 $\rightarrow$ 14.53 dB). Distribution shift between offline oracle labels and recursive online trajectories misallocates microsteps ($K = 2.0$).
- Factorial Confirmation of Distribution Shift:** In an independent 2×3 factorial benchmark crossing selection rules with throttling states, learned utility consistently degrades mean PSNR compared to sensitivity heuristics: $B_2 - B_1 = -2.64$ dB (unthrottled) and $B_5 - B_4 = -2.51$ dB (throttled). This establishes that closed-loop online tracking cannot rely on offline pointwise regressors; map-growth throttling is the true dominant mechanism.
- The Optimal Formulation (OURS):** Combining Coverage + Backlog Throttling with sensitivity selection and fixed $K = 5$ microsteps achieves the peak performance of **19.24 dB** (+3.71 dB over A_0).

4.3 Full-Sequence Benchmark (3,669 Frames)

Table 4 evaluates OURS on the complete 3,669-frame sequence of `tum_fr2_xyz` (122.74 seconds continuous streaming).

- Continuous Stability:** Across 3,669 frames, OURS achieves a final PSNR of **21.08 dB**, SSIM of **0.8847**, and AlexNet LPIPS of **0.5062** at 7.1 FPS, maintaining stable execution without memory leaks or degradation.
- Literature Comparison:** OURS achieves +1.22 dB higher PSNR and +0.205 higher SSIM than Point-

Scene	Policy	Mean PSNR	Final PSNR	SSIM	Depth L1 (m)	Map Size
tum_fr1_xyz	no.op	12.76	11.22	0.7132	0.1428	35,420
	error_only	15.46	15.70	0.8362	0.1175	34,180
	OURS	15.40	15.28	0.8529	0.1042	26,450
	full (Ceiling)	16.27	17.13	0.8757	0.0884	34,890
replica_office0	no.op	20.65	28.70	0.7446	0.5337	69,016
	error_only	27.64	35.46	0.9281	0.4770	66,264
	OURS	28.68	35.60	0.9337	0.4622	56,944
	full (Ceiling)	31.68	40.54	0.9567	0.4151	66,127
tum_fr2_xyz	no.op	13.49	10.07	0.4237	0.1850	35,386
	error_only	17.19	15.75	0.6948	0.1419	30,099
	OURS	18.51	15.03	0.8076	0.1312	28,188
	full (Ceiling)	20.61	20.28	0.8624	0.0953	32,530

Table 5: **Cross-Scene Benchmark on TUM RGB-D and Replica under Matched Budget (15.0 ms).** Demonstrating scene-dependent effects: substantial PSNR surges on `tum_fr2_xyz` (+1.31 dB) and `replica_office0` (+1.03 dB), and enhanced structural SSIM (+2.0%) and depth fidelity (-11.3% L1) on fast handheld motion (`tum_fr1_xyz`).

SLAM while running **36 $\times$ faster** (7.1 vs 0.2 FPS), and +2.67 dB higher PSNR than MonoGS (4.5 $\times$ faster).

4.4 Lifetime Convergence Diagnostic

To understand whether residual perceptual error (LPIPS ≈ 0.50) results from under-optimized primitives, we audited the lifetime optimization history of all 150,000 Gaussians across the entire 3,669-frame sequence:

- High Touch Saturation:** The mean touches per primitive is **220.8**, with a median of **70.0** and 90th percentile of **608.0**.
- Under-Convergence Rejected:** Only **0.96%** of primitives receive < 5 optimization touches (99.04% receive ≥ 5 touches). For early and middle primitives, the fraction with < 5 touches is exactly 0.00% and 0.02%.

This conclusively rejects the hypothesis that primitives suffer from optimization starvation. The remaining perceptual detail is governed by high-frequency view-dependent appearance, as confirmed by the improvement from SH = 0 (LPIPS 0.533) to SH = 1 (LPIPS 0.506).

4.5 Multi-Scene Benchmark

We evaluate cross-scene generalization across real-world TUM RGB-D (`fr1_xyz`, `fr2_xyz`) and synthetic Replica sequences (`office0`) under matched compute budgets ($B = 15.0$ ms) at 320×240 resolution (Table 5):

- Scene-Dependent PSNR Dynamics:** On `tum_fr2_xyz`, OURS delivers **18.51 dB** mean PSNR and 0.8076 SSIM, decisively outperforming compute-matched `error_only` (17.19 dB, 0.6948 SSIM, +**1.31**

Pipeline Stage	Subsystem	Latency	Fraction	Status
Feature Extraction	T_{extract}	0.38 ms	0.33%	CPU/GPU
Fast Attribution (FAA)	T_{attr}	0.85 ms	0.74%	CUDA
Knapsack Ordering	T_{knapsack}	0.77 ms	0.67%	PyTorch
State Synchronization	T_{state}	0.39 ms	0.34%	CPU StateStore
Total Scheduling	$T_{\text{scheduler}}$	2.54 ms	2.22%	Overhead $\ll B_t$
Selective Optimization	T_{opt}	9.25 ms	8.07%	≤ 15.0 ms Budget
Rasterization & Graphs	T_{render}	102.80 ms	89.70%	CUDA <i>gsplat</i>
Total Frame Time	T_{frame}	114.60 ms	100.0%	8.7 FPS (Interactive)

Table 6: **AI Systems Latency Breakdown per Frame (NVIDIA RTX 4050).**

dB, +16.2% SSIM). On *replica_office0*, OURS delivers **28.68 dB** vs 27.64dB (+1.03 dB).

- **Structural and Geometric Fidelity on Fast Motion:** On *tum_fr1_xyz*, characterized by aggressive handheld trajectory displacement, mean photometric PSNR is comparable (15.40 vs 15.46 dB, -0.05 dB), but OURS achieves superior structural fidelity (+2.0% SSIM, 0.8529 vs 0.8362) and significantly lower depth reconstruction error (**-11.3% Depth L1**, 0.1042 vs 0.1175 m).
- **Compact Map Representation:** Across all sequences, OURS maintains strictly bounded primitive memory footprints (26k–57k splats vs 34k–66k), preventing unconstrained primitive explosion.
- **Scientific Finding:** The reconstruction benefit of dual map-growth throttling is **scene-dependent**: the largest raw photometric gains appear on scenes with moderate exploration rates where compute dilution is severe, while highly dynamic motion sequences experience preserved structural SSIM and geometric depth fidelity without map bloat.

5 AI Systems Latency & Resource Breakdown

Table 6 decomposes per-frame wall-clock latency on an NVIDIA GeForce RTX 4050 GPU:

- **Sub-Millisecond Scheduling:** Total scheduling overhead is only **2.54 ms** ($\approx 2.2\%$ of frame time), satisfying real-time constraints.
- **Budget Compliance:** Selective optimization executes in **9.25 ms**, strictly satisfying the 15.0 ms budget.
- **Interactive Frame Rate:** Driven by production ‘*gsplat*’ rasterization and FAA, end-to-end frame processing runs at **8.7 FPS** (114.6 ms), completely replacing legacy unaccelerated Python baselines (7000 ms, 0.14 FPS).
- **OS Memory Guard:** PyTorch memory allocation is capped at 70%, bounding peak VRAM between 210–303 MB with zero desktop compositor stutter.

6 Conclusion

We have presented Adaptive 3DGS, establishing an empirically verified framework for compute-constrained online

RGB-D 3D Gaussian Splatting: (1) Naive error heuristics fail because over 21% of updates induce negative marginal utility, and inter-splat coupling causes extreme non-additivity; (2) Unconstrained densification induces compute dilution, whereas Dual Coverage and Backlog Throttling resolves this bottleneck, delivering +**2.87 dB** mean PSNR gain (paired Cohen’s $d_z = 38.42$, Holm-adjusted $p_{\text{Holm}} \leq 0.0391$, $\eta = 76.4\%$ headroom recovery) with a $3.8\times$ smaller map; (3) Fast Approximate Attribution enables $614\times$ faster attribution (0.85 ms); (4) Validation on 3,669 continuous frames demonstrates sustained interactive mapping (8.7 FPS, 21.08 dB final PSNR); (5) Lifetime convergence diagnostics conclusively reject optimization starvation; and (6) Multi-scene evaluation across TUM RGB-D and Replica confirms scene-dependent advantages with an $\approx 15\times$ speedup over SplatTAM.

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